

Totally Integrated Automation Portal

Software Day8 / PLC_1 [CPU 1215C AC/DC/Rly] / Program blocks

Main [OB1]

Main Properties

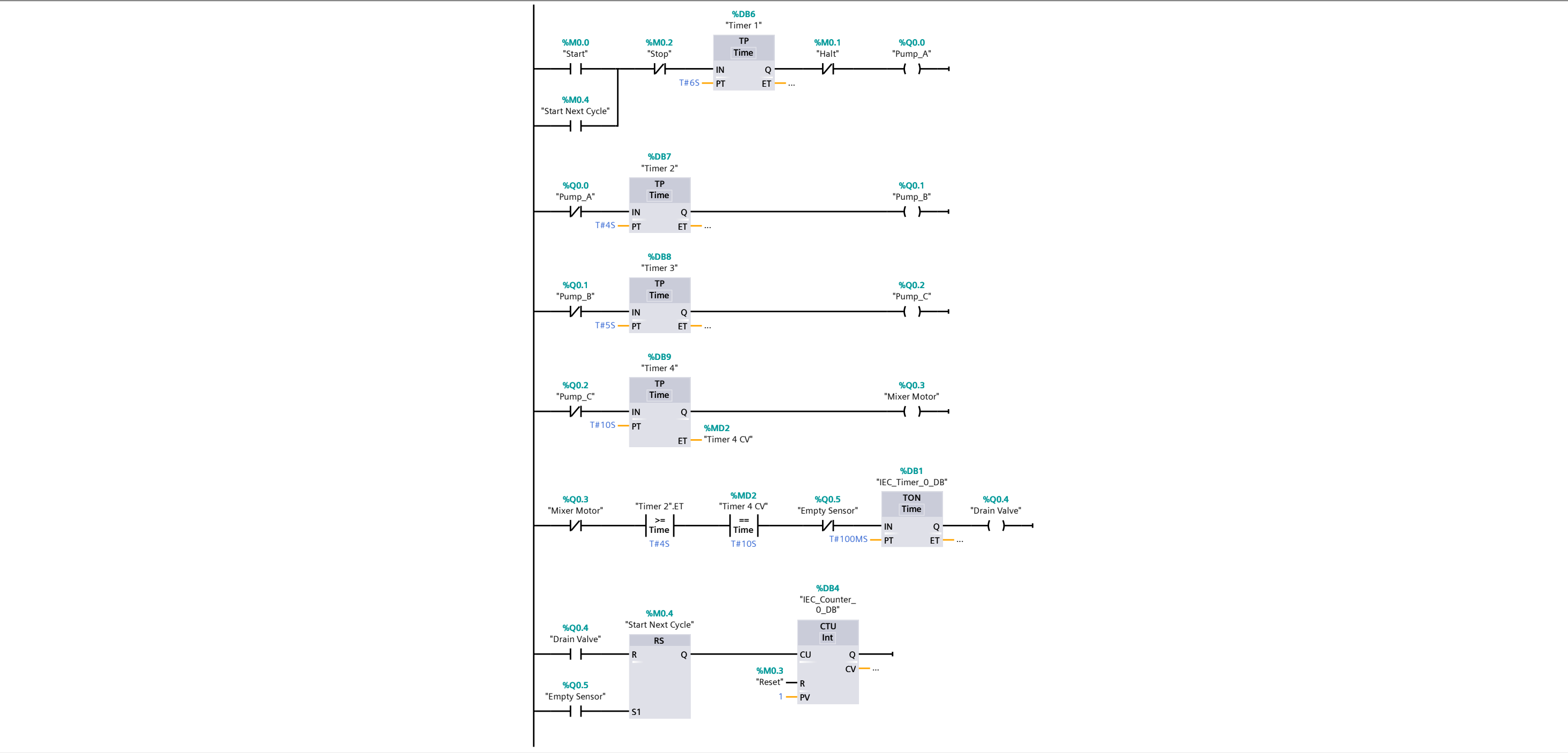
General

Name	Main	Number	1	Type	OB	Language	LAD	Numbering	Automatic
------	------	--------	---	------	----	----------	-----	-----------	-----------

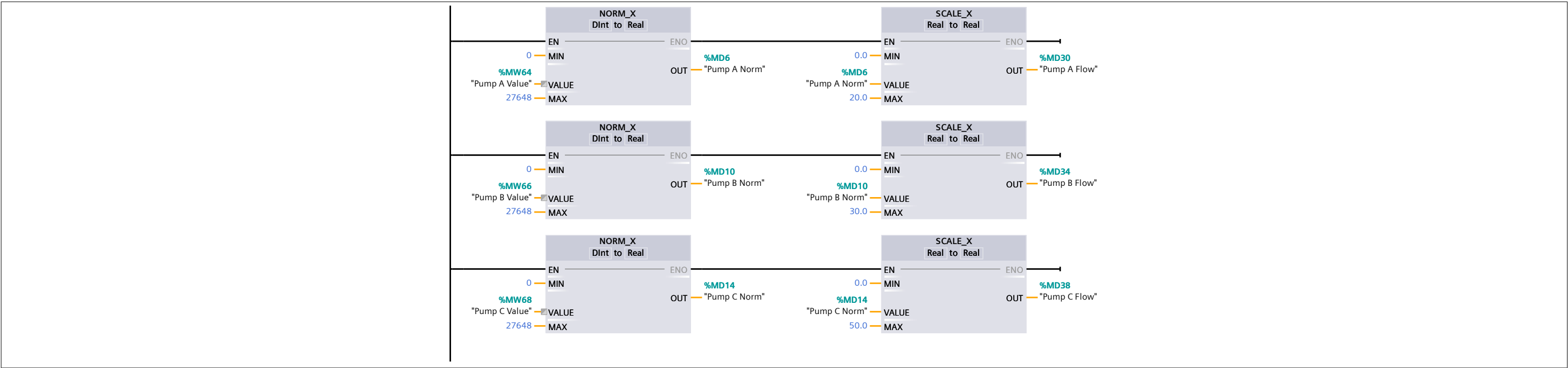
Totally Integrated Automation Portal											
Information											
Title	"Main Program Sweep (Cycle)"	Author		Comment	The "Smart" Chemical Dosing Tank (Difficulty: Advanced - Capstone) Scenario: A tank must mix 3 liquids (A, B, and C) according to a recipe. The PLC controls the pumps using timers. However, the pumps wear out, so the PLC auto-adjusts the timer presets based on a real-time flow rate derived from level sensors. 1. Sequential Filling (Timers): High_Level (At 90% full). Outputs: Pump_A, Pump_B, Pump_C, Mixer_Motor, Drain_Valve. Turn on Pump_A for Preset_A seconds. Stop it. Turn on Pump_B for Preset_B seconds. Stop it. Turn on Pump_C for Preset_C seconds. Stop it. Then run the Mixer Motor for 10 seconds. Finally, open the Drain_Valve until the Empty_Sensor triggers. This is a continuous batch cycle. 2. Batch Counter (Counters): Use a CTU to count how many complete batches (drain cycles) are finished. 3. Adaptive Flow Correction (Math & Comparators): Premise: Pump_A is supposed to deliver 100 Liters. The level sensor at 50% represents 500 Liters. Read the analog Tank_Level during the fill cycle. Calculate the Actual Flow Rate = (Current_Level - Previous_Level) / Elapsed_Time. 4. Use a Comparator. If the Actual Flow Rate is LES than the Expected_Flow_Rate by more than 10%: Safety Interlock (Comparators): If the High_Level sensor triggers before Pump_C finishes its timer, immediately stop all pumps and sound a massive alarm.	Family		Version	0.1	User-defined ID	

Main			
Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

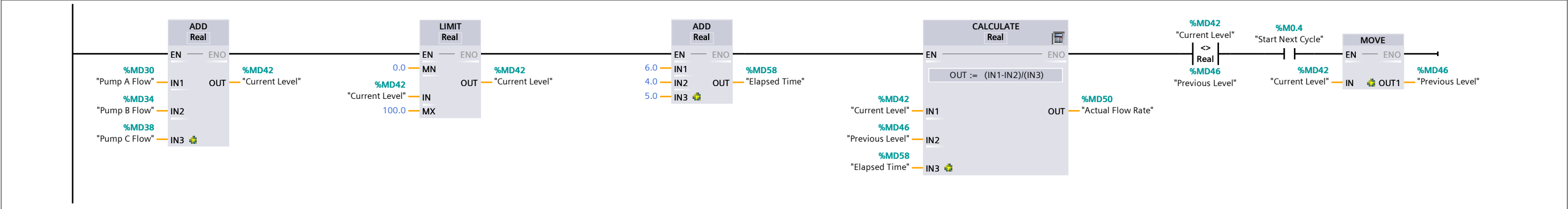
Network 1:



Network 2:



Network 3:



Network 4:

